

Constant Name	Description	Value
MSG_COUNTER_SYNC_TIMEOUT	The maximum amount of time (in milliseconds) which a Node SHALL wait for a <i>MsgCounterSyncRsp</i> after sending a <i>MsgCounterSyncReq</i> .	400 milliseconds

4.12. Message Reliability Protocol (MRP)

The Message Reliability Protocol (MRP) provides confirmation of delivery for messages that require reliability. The protocol is optimized for constrained devices that may not be able to receive a message at the point it is due to be delivered to them. Reliable messaging MAY be enabled on an individual message basis as required by the protocol design of the higher layer application. Reliability is achieved through time-bounded delivery confirmation, ensuring best effort delivery of critical messages over what may be an inherently lossy and unreliable communication medium.

Flow control mechanisms are not incorporated in MRP because it is intended to be used for short interactions with small numbers of messages in them.

4.12.1. Reliable Messaging Header Fields

The following fields are defined in the [Exchange Flags](#) for use exclusively by MRP:

- [R Flag](#)

Indicates a reliable message. This flag SHALL be set by the sender when a message being sent requires the receiver to send back an acknowledgment. To support unreliable messages, this flag bit MAY be clear, so that no acknowledgements are requested from the receiver.

- [A Flag](#)

Indicates the message is acting as an acknowledgement. This flag MAY be set on any message. When set, the *Acknowledged Message Counter* field SHALL be present and valid. This flag SHALL always be set for *MRP Standalone Acknowledgement* messages.

- [Acknowledged Message Counter](#)

This field SHALL be set to the Message Counter of the message that is being acknowledged.

4.12.2. Reliable transfer

When the reliability bit is set, the *reliable message* is transmitted at most [MRP_MAX_TRANSMISSIONS](#) times until an acknowledgement of receipt is received from the peer or a timeout.

4.12.2.1. Retransmissions

Senders provide an automatic retransmission mechanism for reliable messages. In order for the receiver to receive a message reliably, the sender SHALL trigger the automatic retry mechanism after a period of *mrpBackoffTime* milliseconds without receiving an acknowledgement, where *mrpBackoffTime* is calculated according to the formula below. The sender SHALL retry up to a configured maximum number of times ($MRP_MAX_TRANSMISSIONS - 1$) before giving up and notifying